



PROJECT PRESENTATION

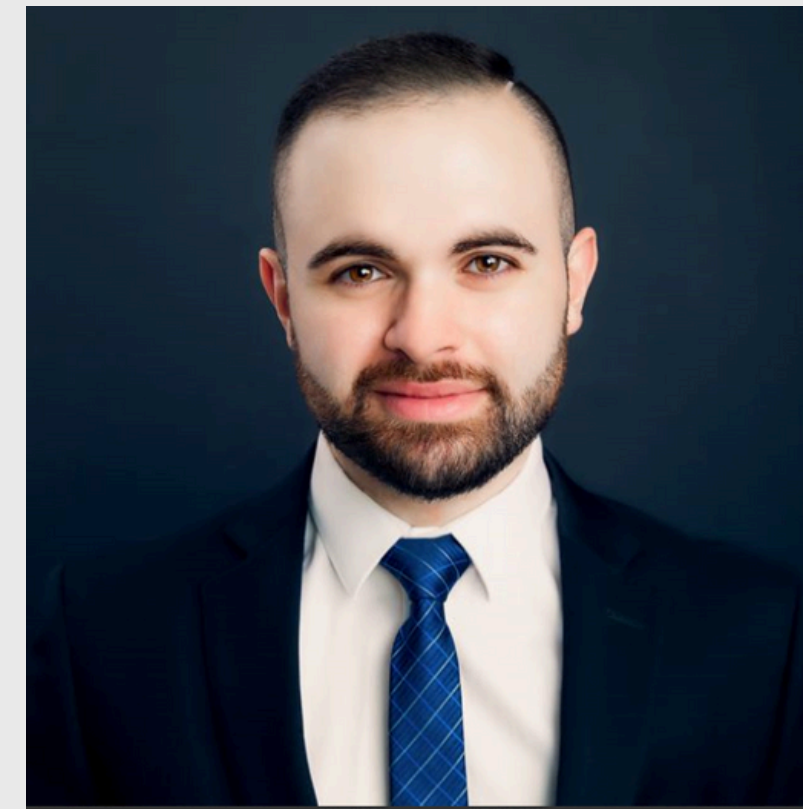
By SustainableEdge

DaRMoD Program

TEAM



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PROJECT OVERVIEW

Problem Statement:

- Small businesses lack accessible and practical ways to obtain ESG ratings, yet these ratings are increasingly critical for sustainable financing and reporting.

Objective:

- Analyze the dataset to explore developing a machine learning model
- Aim to develop a machine learning model that predicts ESG scores for small businesses through simple, easy-to-answer questions.

DATA CONTEXT

Data Source:

- LSEG ESG database – Fortune 500 companies

Dataset Size:

- 500 companies, 60+ features each

Key Features:

- Environmental policies, carbon emissions, employee diversity, board independence, etc.

Preprocessing Steps:

- Feature selection,
- Missing value imputation,
- Normalization

EDA METHODOLOGY

- Feature importance differs sharply by industry.
- Further EDA was performed on the Technology sector (16 tech-related sub-industries: software, IT services, semiconductors, hardware, telecom, etc.).

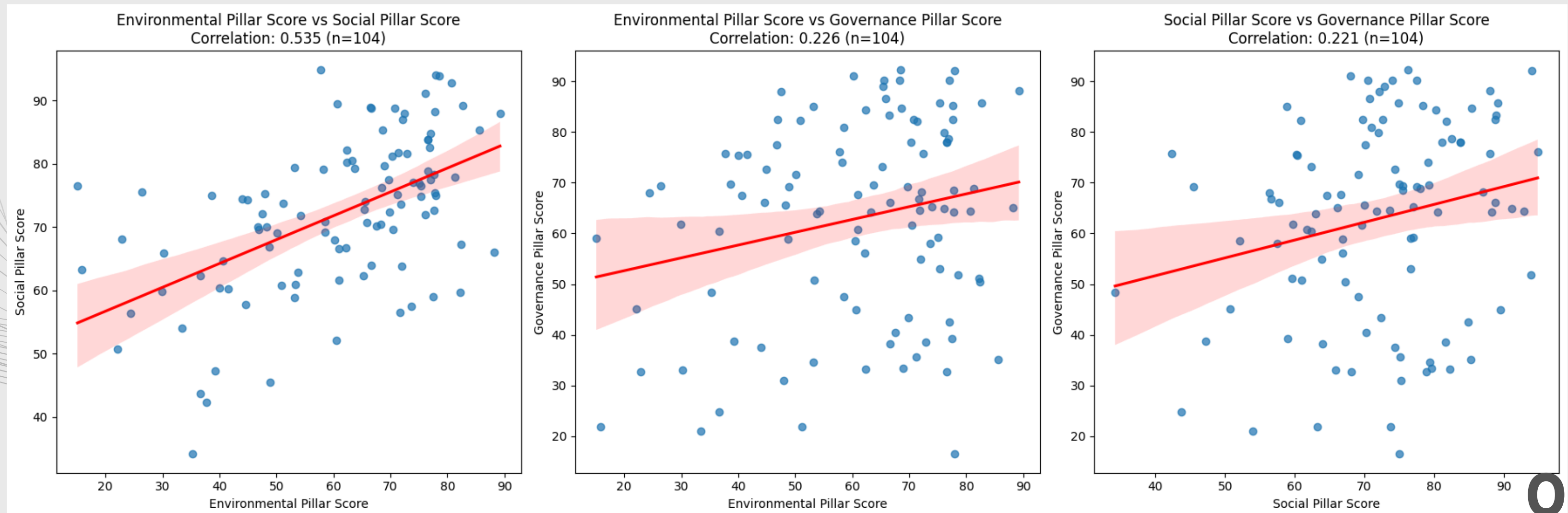
Sector	Total Companies
Technology	162
Other	127
Consumer	48
Industrials	47
Financials	44
Healthcare	40
Energy/Materials	21

EDA METHODOLOGY

- Correlation Analysis of ESG Components: Examined relationships between ESG Score and its Environmental, Social, and Governance pillars
- Inter-Pillar Relationships: Explored connections between the three ESG pillars to understand integration
- Resource Efficiency Analysis: Investigated correlations between operational metrics (energy, water, waste efficiency)
- ESG Drivers Investigation: Identified key predictors of ESG scores, focusing on non-circular relationships
- Gap Analysis: Discovered disconnect between ESG ratings and actual resource efficiency

TECH SECTOR EDA RESULTS

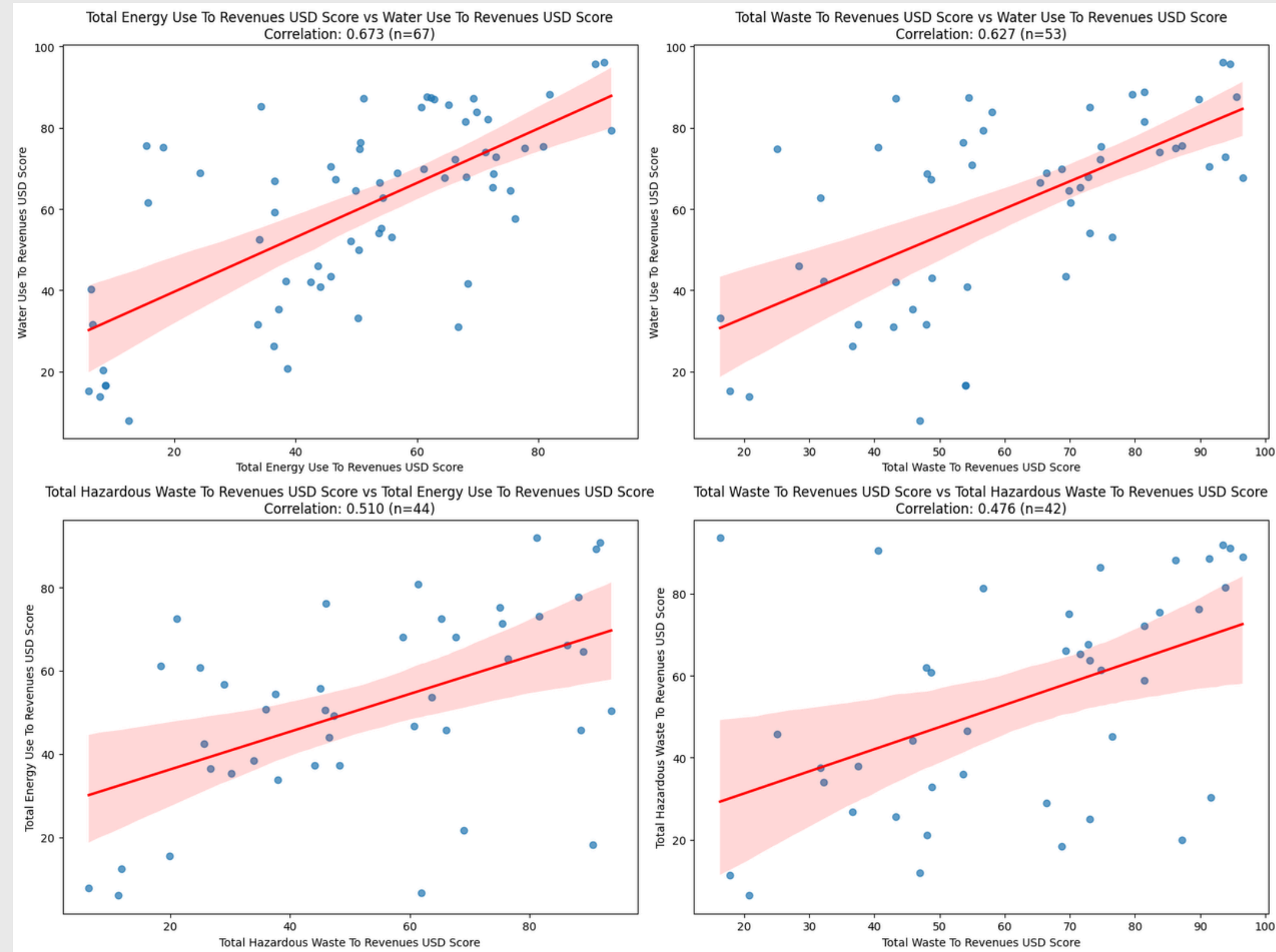
- Governance has strongest influence on overall ESG ($r=0.779$), followed by Social ($r=0.712$) and Environmental ($r=0.657$)
- Environmental and Social performance correlate ($r=0.535$) while Governance develops independently



TECH SECTOR EDA RESULTS

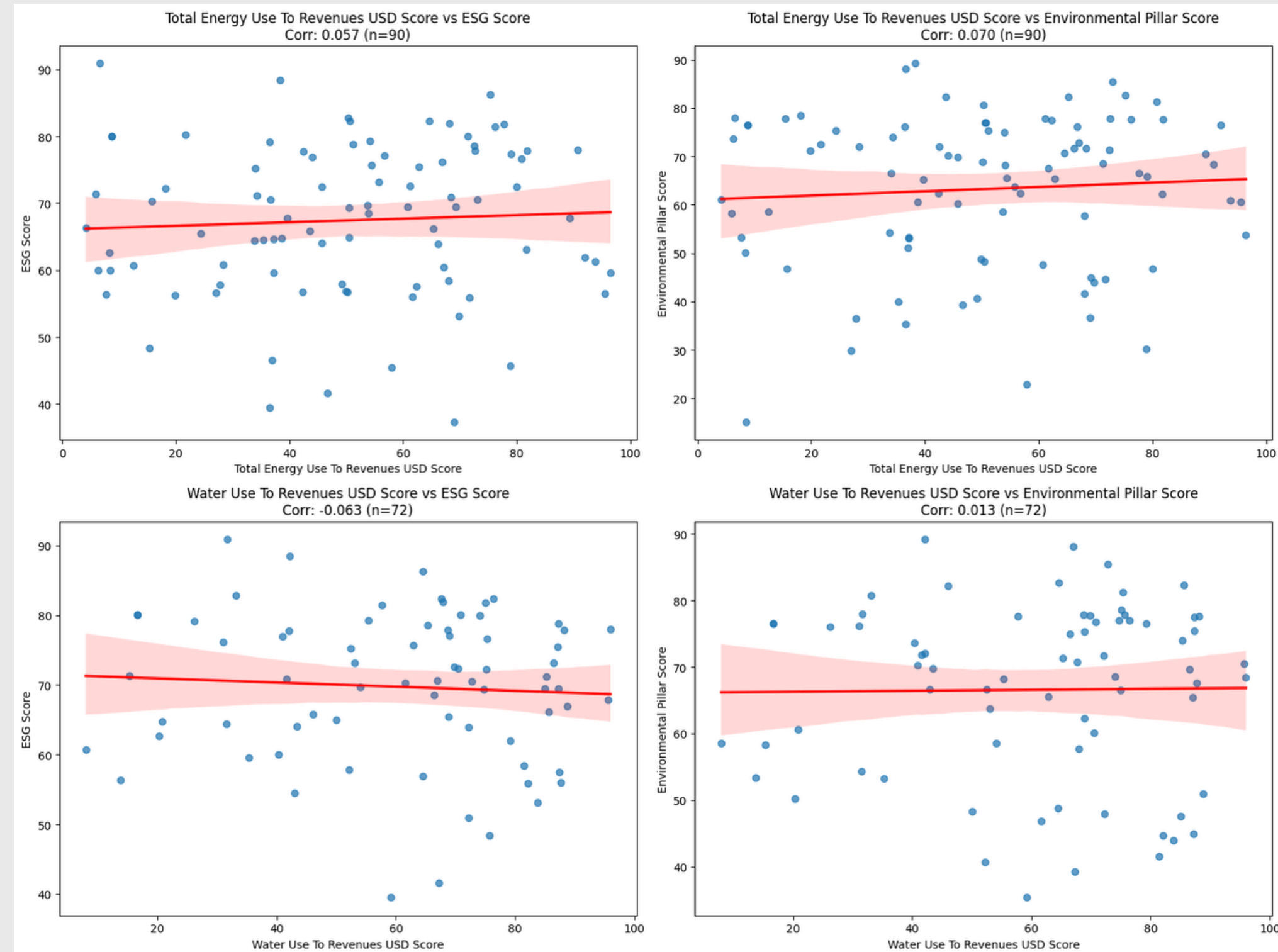
- Resource efficiency metrics show strong inter-relationships ($r \approx 0.5-0.7$).

Companies good at managing one resource, tend to be good at managing others.



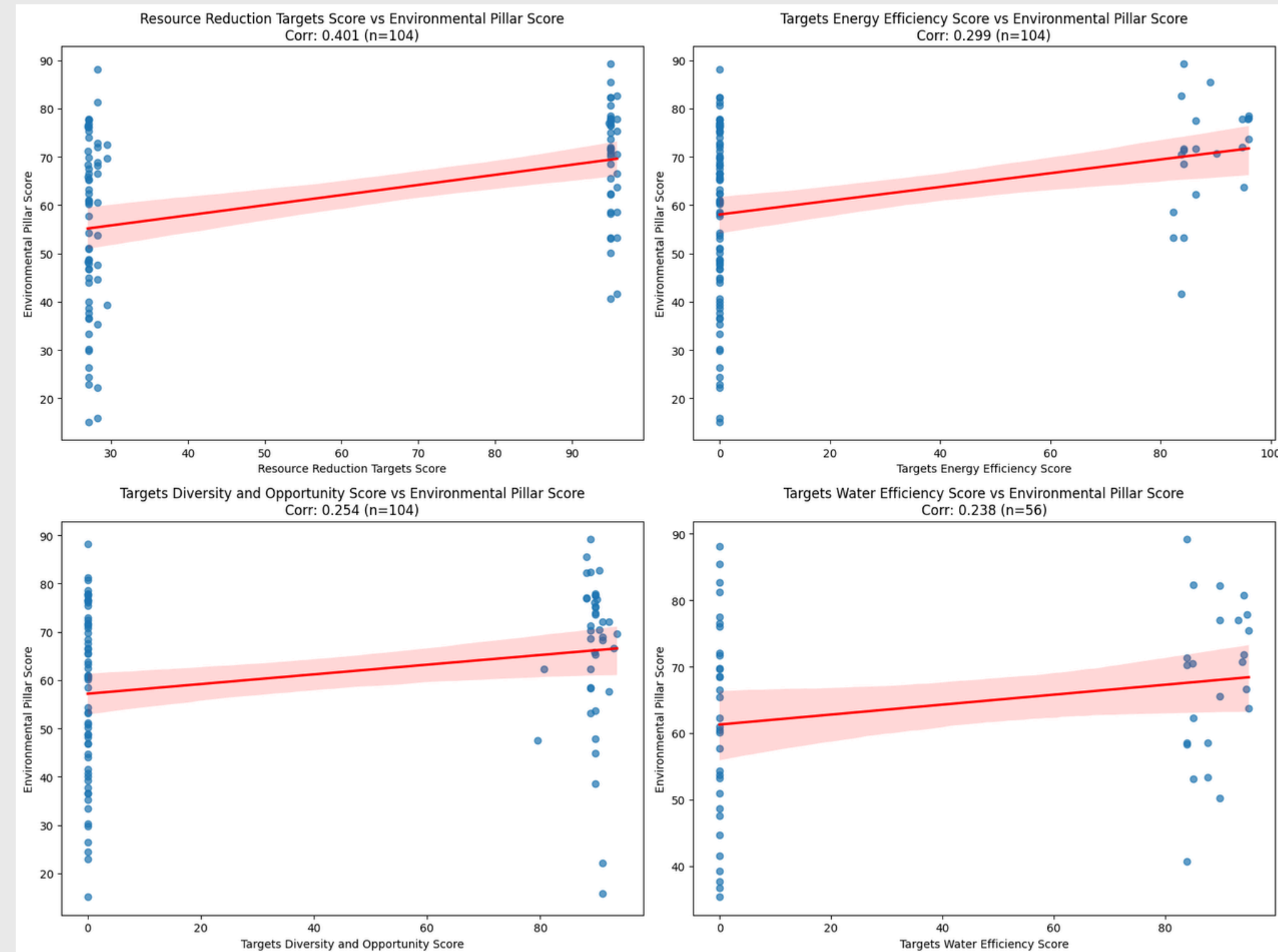
TECH SECTOR EDA RESULTS

- Targets and policies predict ESG scores far better than actual performance metrics surprisingly weak correlation between environmental ratings and actual resource efficiency ($r < 0.1$)



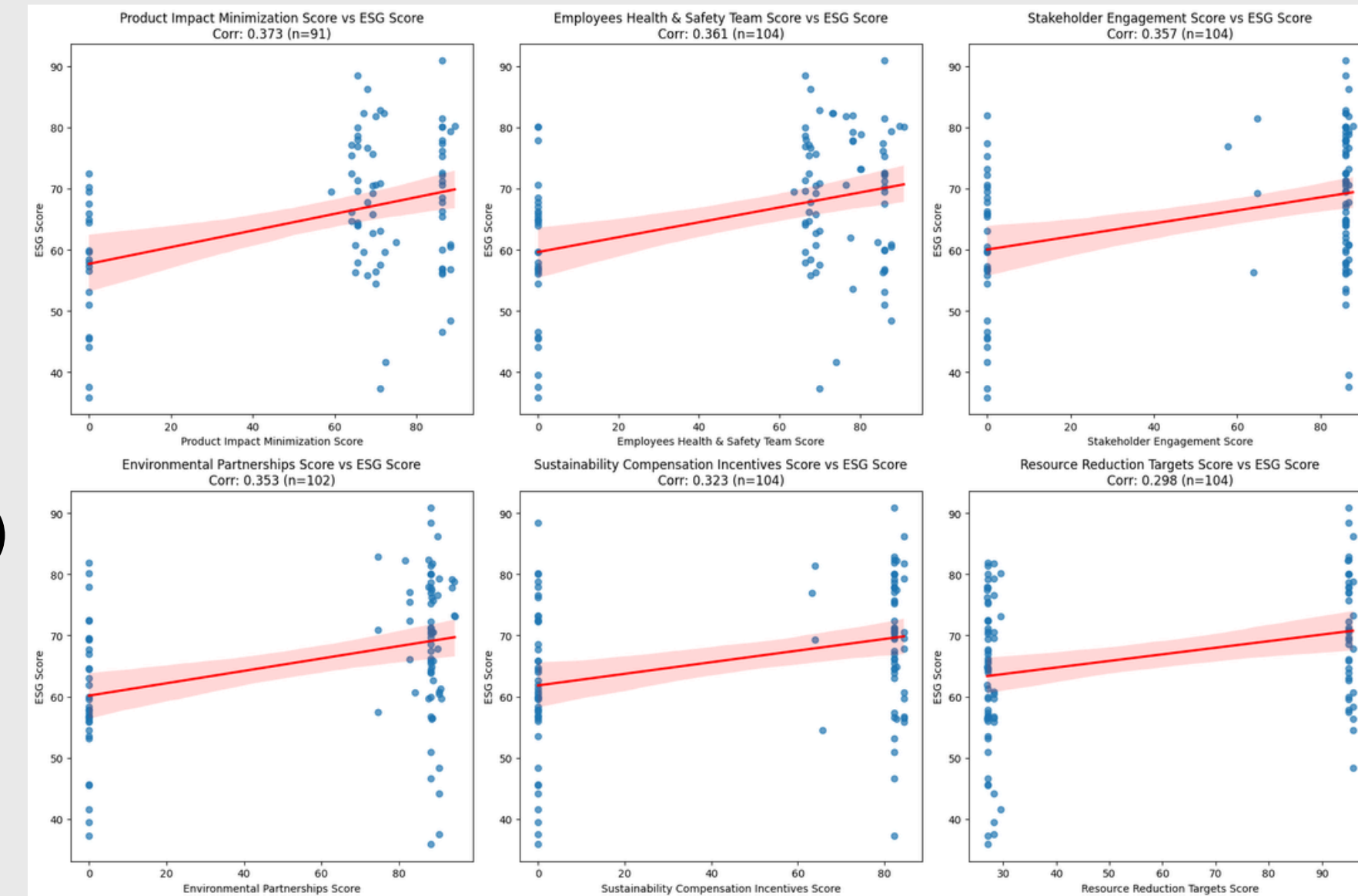
TECH SECTOR EDA RESULTS

- This analysis reveals that ESG ratings in tech reward commitment and governance structures more than operational sustainability performance.



TOP PREDICTORS

- Product Impact Minimization Score ($r=0.373$)
- Employees Health & Safety Score ($r=0.361$)
- Stakeholder Engagement Score ($r=0.357$)
- Environmental Partnerships Score ($r=0.353$)
- Sustainability Compensation Incentives ($r=0.323$)
- Resource Reduction Targets Score ($r=0.298$)
- Quality Management Systems Score ($r=0.282$)
- Targets Diversity and Opportunity Score ($r=0.269$)
- Targets Water Efficiency Score ($r=0.259$)



CHALLENGES

- Weak predictive power – best feature only explains ~14% of ESG score variance.
- Performance data weakly linked – outcomes (e.g., energy use) have low correlation.
- No insight into startup internal decision-making or execution quality.
- Static data – no view of change, improvement, or momentum over time.
- Large company bias – current dataset (S&P 500) doesn't reflect startup realities.
- Missing validation – we lack data for startups to benchmark our model.
- Limited data access – bulk downloading is not allowed and poses compliance risks.
- More research needed – data provider doesn't disclose ESG formula weights, but others do with different features. Modeling is redundant if a published formula already exists.

NEXT STEPS

- Download more tech industry data
- Analyze correlation between company size and ESG score
- Search for published ESG scoring formulas as some providers may disclose their methodology and data.
- Reach out to startups for insights on how they approach ESG in practice.
- Combine top predictors to build an ensemble model



THANK YOU